

Integration of Wearable Health Device Data and Retinal Imaging for Multimodal Systemic Disease Stratification

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Abstract

Wearable health devices provide continuous, real-time insights into physiological patterns that may reflect underlying systemic diseases. However, validating these signals often requires invasive and costly clinical tests. As the only externally visible microvascular structure, the retina offers a unique noninvasive window into vascular health. Retinal biomarkers including vessel calibers (CRAE, CRVE), arteriovenous ratio (AVR), vessel density, fractal dimension, and tortuosity have been associated with systemic conditions such as diabetes, hypertension, and cardiovascular disease. In this study, we leveraged the AIREADI dataset to investigate how retinal vascular features vary across stratified levels of physiological dysregulation, as measured by wearable-derived metrics (e.g., heart rate, SpO_2 , sleep, activity, and stress scores) collected over 10 days. Participants were categorized into low, moderate, and severe physiological severity groups and further stratified by six chronic disease conditions. Our analysis revealed significant associations between higher physiological severity and venular dilation (increased CRVE), reduced arteriovenous ratio (AVR), and elevated venous fractal dimension. These trends were consistent across several disease-specific subgroups, particularly in diabetes, hypertension, and kidney disease. However, parameters such as CRAE and tortuosity showed no consistent or statistically significant changes. These findings support the complementary role of retinal imaging and wearable sensors in passive, scalable monitoring of systemic vascular health, offering promise for early risk stratification and precision health interventions.

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CCS Concepts

• **Human-centered computing** → Empirical studies in ubiquitous and mobile computing; • **Applied computing** → Health care information systems.

Keywords

Oculomics, AI for Digital Health, Wearable Health Data, Fitness Trackers, Multimodal Analysis, Early Disease Diagnosis, Systemic Health Monitoring

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1 Introduction

Wearable health technologies like smartwatches and fitness trackers enable continuous and non-invasive monitoring of physiological parameters such as heart rate, peripheral oxygen saturation (SpO_2), respiratory rate, sleep quality, and physical activity [2, 27]. These high frequency data streams offer valuable insights into individual health trajectories and support early detection of systemic dysfunctions including cardiovascular, metabolic, and neurological conditions [31, 32]. However, wearable derived data can be affected by signal noise and variability in user compliance. Moreover, translating short-term physiological fluctuations into reliable indicators of long-term disease risk remains challenging, thus requiring clinical validation through conventional diagnostic approaches that are often invasive, immobile, and expensive. Parallely advances in color fundus photography have led to the emergence of oculomics, a field which uses retinal imaging to identify systemic diseases through ocular biomarkers [37]. As an accessible extension of the central nervous and vascular systems, the retina provides a unique noninvasive view of cumulative vascular health. Changes in vessel caliber,

shape, and density have been linked to conditions such as diabetes, hypertension, kidney and heart disease, and neurodegenerative disorders [17, 24, 40]. Modern retinal analytics tools such as SIVA and AutoMorph use deep learning to quantify features like central retinal arteriolar equivalent (CRAE), central retinal venular equivalent (CRVE), arteriovenous ratio (AVR), fractal dimension, and vessel density based on standardized formulas [18, 46], improving consistency and reducing manual bias in large-scale research. The integration of wearable data with retinal imaging offers a promising multimodal framework for systemic health assessment [38]. While wearables capture real-time dynamic physiological fluctuations, retinal images reflect more stable vascular features that accumulate over time. This combination enhances subclinical disease detection, supports risk stratification, and enables personalized digital phenotyping. Large population based cohorts such as the UK Biobank and AIREADI now allow for the integration of physiological, ocular, and clinical data. Despite this potential, the relationship between physical activity patterns and retinal microvascular features remains insufficiently studied, particularly in diverse community-based populations.

This study aims to address this gap by using data from the AIREADI cohort and integrating wearable derived activity metrics with retinal biomarkers. Among 1,067 participants in the dataset, 905 had at least 10 days of valid recordings across six Garmin Vivosmart 5 sensors. These readings were used to derive nine composite lifestyle parameters, each labeled as normal or abnormal on a daily basis, and collapsed into participant-level binary scores. Their summation yielded a composite abnormality score which was further used to categorize individuals into low, moderate, and severe groups. Concurrently, fundus images from the Topcon Maestro2 device were processed using a modified AutoMorph pipeline to extract vascular features. Analysis of variance and linear trend tests were used to assess feature changes across severity groups, with further stratification by six conditions: heart disease, kidney disease, neurological disorders, hypertension, type 2 diabetes, and hyperlipidemia. We hypothesize that individuals in the moderate and severe categories will show narrower arterioles, wider venules, lower AVR, reduced fractal dimension, and lower vessel density, particularly in those with specific clinical conditions. These associations would support the use of retinal imaging and wearable data as complementary tools for systemic health monitoring and risk mitigation. The major contributions of this study could be summarized as follows:

- This study investigates the relationship between retinal vascular biomarkers and physiological metrics from nine wearable sensors, using real-world data from a large, diverse cohort.
- We analyze how retinal biomarkers vary across systemic disease groups stratified by wearable-measured activity levels, highlighting the impact of habitual physical activity on systemic disease progression and ocular vascular health.

2 Related Works

2.1 Oculomics

The retina provides a unique, noninvasive window into systemic vascular health. Landmark epidemiological studies such as ARIC

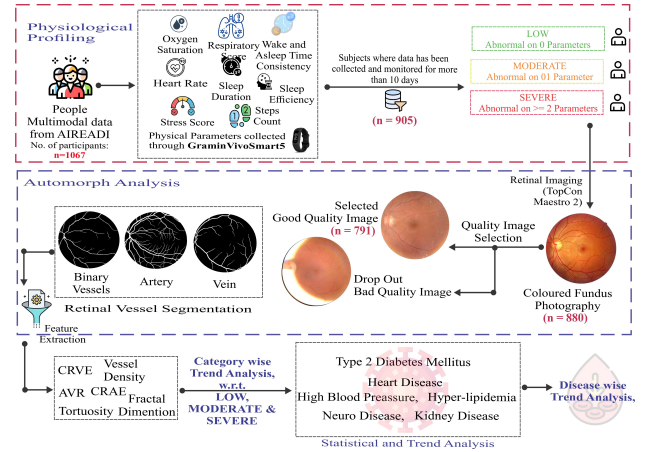


Figure 1: A schematic overview of the proposed three-stage analysis pipeline, outlining the process of physiological profiling, retinal vascular feature extraction, and disease-wise trend evaluation

[23] and MESA [8] have consistently demonstrated associations between retinal vascular characteristics and systemic conditions including hypertension, diabetes, and cardiovascular diseases. Foundational work by Wong *et al.* [41] established that arteriolar narrowing predicts future hypertension, while venular dilation signals inflammatory burden and elevated cardiovascular risk. Beyond vessel calibers, a growing body of research has highlighted advanced vascular features such as tortuosity, fractal dimension, and AVR as informative biomarkers for systemic diseases. These markers have been linked to chronic kidney disease [42], type 2 diabetes [29], cardiovascular events [12], and neurological conditions [19]. For instance, a reduction in fractal dimension has been associated with neurodegeneration, increased stroke risk, and mortality [20, 21], while elevated tortuosity and altered AVR have been implicated in early vascular dysfunction and genetic susceptibility to coronary artery disease [30, 36].

Despite substantial advancements in vessel segmentation, many early computational methods ranging from unsupervised graph-based algorithms [15, 33] to supervised deep learning models [28] primarily addressed isolated tasks such as binary vessel segmentation or artery–vein classification. These approaches often lacked comprehensive preprocessing or downstream feature quantification capabilities [10, 25, 44], limiting their clinical utility. Additionally, semi-automated tools such as RA, IVAN [5], and VAMPIRE [26] frequently required manual correction, introducing subjective variability and constraining scalability [11, 39]. Compounding these challenges were issues related to poor generalizability across imaging platforms and patient populations, as well as the absence of integrated image quality control mechanisms [13, 22]. To address these issues, Zhou *et al.* proposed AutoMorph [46], a fully automated pipeline that integrates image quality grading, segmentation, and vascular feature extraction. Studies using AutoMorph have reported high reproducibility in extracting features such as fractal dimension, vessel density, AVR, and tortuosity. Engelmann *et al.* [9] and Giesser *et al.* [14] validated the stability of these features for clinical modeling, while Choi *et al.* [3] applied them in

pediatric ADHD screening. Integration with foundation models like RETFound [45] further enhances the generalizability of these tools across diverse disease contexts, enabling scalable and robust oculomic analyses.

2.2 Wearables

Wearable biosensors enable continuous monitoring of physiological parameters such as heart rate, sleep, and physical activity [16]. Large-scale studies like the UK Biobank [6] and Apple Heart and Movement Study (AHMS) [35] have demonstrated the feasibility of collecting such data at scale, supporting investigations into health and behavior over extended periods. These data have been used to quantify clinical phenotypes, such as motor fluctuations in Parkinson’s disease [4], highlighting the potential of wearables for real-time, objective monitoring. However, challenges persist due to noise, variability, and incomplete data, often linked to device adherence [34]. Recent advances in self-supervised learning have addressed these limitations by enabling generalizable models trained on large unlabeled datasets. Yuan *et al.* [43] developed activity recognition models from over 700,000 person-days of accelerometer data, while Abbaspourazad *et al.* [1] trained modality-agnostic models on PPG and ECG signals from AHMS, capturing features relevant to cardiometabolic health. These efforts underscore the growing utility of wearables, paired with scalable modeling approaches, for robust physiological monitoring.

3 Methodology

Our approach follows a three-stage pipeline comprising of (1) construction of individual physiological profiles using minute-level wearable data from Garmin Vivosmart5, (2) vascular feature extraction from retinal fundus images captured via Topcon Maestro2 using the AutoMorph pipeline, and (3) physiological severity wise trend analysis across various health conditions. The physiological profiles include clinically informed lifestyle risk indicators computed over a minimum 10 day window. Retinal biomarkers span 10 vascular traits reflecting microvascular health. Statistical analyses, including one-way ANOVA and linear trend tests, were employed to assess variations across disease subgroups. The overall methodology is illustrated in Fig. 1 and discussed in detail in the following subsections.

Table 1: Physiological Parameters and Abnormality Criteria

Parameter	Definition	Abnormality Criteria	Method
Heart Rate	Avg. beats/min	<50 or >100 bpm	Fixed
SpO ₂	Avg. O ₂ during sleep	<90% for >10% sleep	Fixed
Step Count	Steps/day (activity)	≥50% drop from 7-d avg.	Baseline
Sleep Duration	Sleep per night (hrs)	<4 hrs	Fixed
Sleep Efficiency	Sleep time/Time in bed (%)	<70%	Fixed
Wake Time Variability	Wake time deviation	>2 hr from 7-d avg.	Baseline
Sleep Time Variability	Sleep onset deviation	>2 hr from 7-d avg.	Baseline
Stress Score	Garmin stress index	>70	Fixed
Respiratory Rate	Breaths/min at rest	<10 or >22 bpm	Fixed

3.1 Dataset

This study leverages data from the Artificial Intelligence Ready and Exploratory Atlas for Diabetes Insights (AI-READI) consortium,

which includes phenotypic profiles of 4,000 participants across three academic centers. For our analysis, we used currently released data from 1,067 participants, focusing on retinal CFPs and wearable derived physiological signals. To maintain imaging consistency, only CFPs captured using the Topcon Maestro2 device were included. Physiological data were obtained from Garmin Vivosmart 5 smart-watches, providing minute-level measurements over at least 10 days in free-living conditions. This integration of ocular and wearable modalities enables scalable exploration of systemic physiology via the retinal microvasculature, supported by the demographic diversity of the cohort.

Table 2: Distribution of patients with various systemic diseases across levels of Physiological Severity defined using wearable data

Category (N = No. of subjects)	Low (N = 313)	Moderate (N = 293)	Severe (N = 185)	Total (N = 791)
Mean Age (in years)	60.3	60.5	58.0	59.8
Kidney Disease	26	28	21	75
Heart Disease	66	67	48	181
Neuro Disease	67	71	28	166
Type2 Diabetes	94	110	99	303
Hypertension	142	162	115	419
Hyperlipidemia	149	150	95	394

3.2 Physiological Profile Creation

Physiological sensor data were available for 905 of the 1,067 participants, capturing key parameters such as heart rate, oxygen saturation, step count, sleep duration, stress score, and respiratory rate directly from wearable devices. Secondary metrics including sleep efficiency (ratio of sleep duration to total time in bed) and sleep consistency (regularity of sleep-wake patterns) were also derived to enrich the physiological profile. To reduce noise from real-world uncontrolled settings, minute-level data were averaged daily, emphasizing stable patterns. For analytical rigor, only participants with at least 10 complete days of data across all sensors were included, resulting in a final cohort of 880 individuals.

Clinically validated thresholds were used to identify abnormalities within each parameter. Fixed thresholds were applied to most variables, such as heart rate and oxygen saturation, while personalized baselines, calculated from each participant’s 7-day average were used for parameters with high inter-individual variability, like sleep-wake timing and step count. Table 1 details the parameters, thresholding methods, and criteria for abnormality. Daily binary labels (0 for normal, 1 for abnormal) were assigned based on these thresholds, and to minimize short-term fluctuations, the mode of these labels was computed across the entire observation window to capture persistent physiological trends.

An aggregate abnormality score was then calculated by summing the final binary labels across all parameters, providing a composite index of physiological dysregulation. Participants were stratified into three risk categories: low risk (score 0), moderate abnormality (score 1), and severe abnormality (score greater than 2). This composite abnormality score has been used as an indicator of physiological severity. When discussing severity within individual systemic disease groups (e.g., diabetes, hypertension), we refer exclusively to physiological severity derived from wearable data, not to clinical or

systemic disease severity. This distinction is maintained throughout the manuscript. To further explore links between physiological dysregulation and chronic disease, participants within each risk group were categorized based on six conditions—kidney disease, heart disease, neurological disorders, type 2 diabetes mellitus, hypertension, and hyperlipidemia—determined through self-report and clinical validation. Table 2 summarizes participant distribution across abnormality groups and disease categories.

3.3 Vascular Feature Extraction

The dataset consists of retinal fundus images acquired using the Topcon Maestro2 device under various protocols, including disc-centered and macula-centered views, and multiple spatial resolutions such as 3D wide CFP and 3D macula CFP. For this study, we selected the images captured using 3D wide CFP protocol due to its widest field of view, which captures the most comprehensive retinal anatomical features.

All images were processed using an enhanced version of the AutoMorph pipeline—an open-source framework for automated retinal image analysis. This pipeline includes four key modules: preprocessing, image quality control, anatomical segmentation, and retinal feature quantification. Preprocessing was applied in the form of affine transformations (centering, scaling, cropping) to standardize and spatially align images. Quality control employed an EfficientNet-B4 model trained on EyePACS-Q data, with confidence-based thresholding to exclude low-quality images, ensuring only gradable images proceeded. From the 880 participants with wearable data, 791 had at least one high-quality fundus image suitable for analysis. The anatomical segmentation module used specialized deep learning models for distinct tasks: LW-Net for binary vessel segmentation preserving fine vascular details; a multi-class network for artery-vein segmentation based on structural and spatial cues; and a U-Net leveraging circular symmetry priors for accurate optic disc segmentation. In the final stage, vascular morphology biomarkers were extracted from these segmentations, including classical ophthalmic features such as CRAE, CRVE, AVR, fractal dimension, tortuosity, and vessel density. Standardized methods like Knudtson and Hubbard formulas were used for calculation of these parameters to ensure clinical relevance and consistency. This study focuses on 10 key parameters—CRAE Knudtson (zone c), CRVE Knudtson (zone c), AVR Knudtson (zone c), fractal dimension (zone c), vessel density (zone c), artery fractal dimension, vein fractal dimension, artery distance tortuosity (zone c), vein distance tortuosity (zone c), and distance tortuosity (zone c)—all of which are well-established markers for systemic disease classification and have shown strong performance in prior AutoMorph studies [7].

3.4 Statistical Analysis

To examine differences in retinal vascular parameters across physiological risk categories, one-way ANOVA was performed to test for significant variations in mean values among the low, moderate, and severe abnormality groups. This analysis identified features exhibiting meaningful inter-group differences. Complementing this, linear trend analysis assessed whether parameter values showed a consistent increase or decrease corresponding to escalating physiological dysregulation. Using a regression-based approach with risk

categories treated as ordinal levels, the direction and significance of these trends were quantified.

In addition to analyzing the overall cohort, these statistical tests were applied within six disease specific subgroups, stratified earlier. This stratified approach allowed evaluation of whether the observed patterns persisted among individuals with particular chronic conditions. Statistical significance was defined at a threshold of $p < 0.05$. Parameters demonstrating both significant group differences and consistent linear trends were considered as promising systemic health biomarkers for further study.

4 Results and Discussion

The retinal vascular features exhibited statistically significant changes across physiological severity categories, reflecting systemic microvascular alterations associated with increasing disease burden. These changes were consistently supported by both one-way ANOVA tests assessing mean differences and linear trend analyses that capture the directional progression of vascular remodeling with advancing severity.

Table 3: Trends in Retinal Features Across Different Severity Groups[†]

Retinal Feature	Low	Moderate	Severe	ANOVA p-value	p-trend
CRAE	185.504 ± 23.418	186.073 ± 20.788	188.996 ± 19.858	0.231	0.117
CRVE	299.207 ± 34.777	304.200 ± 34.673	311.346 ± 32.798	0.001	0.000
AVR	0.623 ± 0.057	0.616 ± 0.060	0.611 ± 0.061	0.105	0.035
FD	1.347 ± 0.040	1.350 ± 0.041	1.354 ± 0.039	0.166	0.059
VD	0.031 ± 0.007	0.032 ± 0.007	0.033 ± 0.007	0.085	0.034
Artery FD	1.301 ± 0.035	1.300 ± 0.036	1.300 ± 0.031	0.963	0.796
Vein FD	1.325 ± 0.033	1.328 ± 0.034	1.335 ± 0.028	0.005	0.002
Artery DT	3.893 ± 2.055	4.066 ± 2.030	4.126 ± 2.131	0.442	0.219
Vein DT	3.673 ± 1.727	3.549 ± 1.756	3.717 ± 1.680	0.561	0.922
DT	3.209 ± 1.068	3.229 ± 1.008	3.335 ± 1.001	0.430	0.234

In the general physiological severity cohort, CRVE increased significantly from 299.2 μm (low) to 311.4 μm (severe), with a strong trend ($p\text{-trend} = 0.000$), indicating progressive venular dilation likely due to systemic inflammation and microvascular dysfunction (Table 3). CRAE remained stable, rising slightly from 185.5 μm to 189.0 μm , suggesting minimal arteriolar changes. The AVR declined subtly but significantly from 0.62 to 0.61 ($p\text{-trend} = 0.035$), reflecting venular enlargement relative to arterioles and microvascular imbalance. Overall fractal dimension showed a mild, non-significant increase, while vein FD rose notably from 1.32 to 1.33 ($p\text{-trend} = 0.002$), indicating venular network remodeling with increasing severity. Vessel density increased slightly from 0.031 to 0.033 ($p\text{-trend} = 0.034$), suggesting early compensatory angiogenesis or microvascular adaptation. Arterial FD remained unchanged, indicating preserved arterial branching. Tortuosity measures showed minor, non-significant increases, reflecting limited vessel twisting changes. These findings highlight venular dilation and remodeling as primary retinal microvascular responses in systemic disease progression, accompanied by subtle increases in vascular complexity and density, while arterial architecture remains relatively stable. This pattern underscores ongoing microvascular stress and inflammation that may impair tissue perfusion and contribute to systemic vascular dysfunction.

Retinal vascular profiles in type 2 diabetes showed selective but meaningful deviations with increasing severity. CRVE rose consistently from 296.9 μm to 311.2 μm , marking venular dilation likely triggered by cumulative microvascular stress from prolonged hyperglycemia. CRAE, however, remained mostly stable across the low and moderate groups, with a minor uptick in the severe stage suggesting limited arteriolar involvement until later progression. AVR showed a clear decline from low (0.627) to moderate (0.609) severity, with a slight recovery in the severe group (0.611), pointing to a disrupted arterio-venous balance in the mid stages of disease. Retinal complexity also shifted subtly, with FD rising from 1.34 to 1.36 and vein-specific FD increasing significantly, highlighting early venular remodeling. VD followed a similar upward trend, most pronounced in the severe group, possibly representing compensatory capillary responses. Despite fluctuations, arterial and venous tortuosity did not exhibit a consistent pattern, suggesting that vessel curvature may not be a defining marker at this stage. These findings suggest that early microvascular remodeling in diabetes, particularly venular dilation and complexity changes, reflects escalating systemic dysregulation (Table 4).

Table 4: Trends in Retinal Features Across Different Severity Groups Stratified by Presence of Type 2 Diabetes[†]

Retinal Feature	Low	Moderate	Severe	ANOVA p-value	p-trend
CRAE	185.227 $\pm$ 24.560	185.211 $\pm$ 20.630	188.836 $\pm$ 16.672	0.390	0.237
CRVE	296.920 $\pm$ 36.555	306.246 $\pm$ 34.379	311.152 $\pm$ 30.488	0.017	0.005
AVR	0.627 $\pm$ 0.060	0.609 $\pm$ 0.064	0.611 $\pm$ 0.059	0.105	0.091
FD	1.342 $\pm$ 0.041	1.345 $\pm$ 0.041	1.356 $\pm$ 0.039	0.037	0.016
VD	0.031 $\pm$ 0.007	0.031 $\pm$ 0.007	0.033 $\pm$ 0.007	0.062	0.045
Artery FD	1.291 $\pm$ 0.040	1.293 $\pm$ 0.036	1.299 $\pm$ 0.030	0.264	0.108
Vein FD	1.320 $\pm$ 0.035	1.320 $\pm$ 0.035	1.335 $\pm$ 0.029	0.003	0.003
Artery DT	3.972 $\pm$ 2.175	4.417 $\pm$ 2.281	4.141 $\pm$ 2.165	0.381	0.620
Vein DT	3.751 $\pm$ 1.833	3.601 $\pm$ 1.686	3.847 $\pm$ 1.838	0.631	0.707
DT	3.299 $\pm$ 1.177	3.371 $\pm$ 1.098	3.301 $\pm$ 0.984	0.874	0.997

In individuals with hypertension, the retinal vasculature exhibited signs of cumulative vascular stress, most notably through progressive venular dilation. CRVE values rose consistently from 297.4 μm in the low group to 312.1 μm in severe cases (p-trend = 0.001), indicative of chronic venous congestion. However, CRAE showed only mild, irregular increases, lacking a clear severity-linked gradient, suggesting that arteriolar responses are less uniform. Microvascular architecture also shifted, as fractal dimension increased significantly from 1.34 to 1.36 (p-trend = 0.007), accompanied by a small rise in vessel density, changes that may reflect early-stage vascular remodeling or compensatory neovascularization. Vein FD also rose significantly, indicating venular structural adaptation. Vessel tortuosity exhibited mixed trends: arterial DT showed a gradual, non-significant increase (3.72 to 4.07), whereas vein tortuosity dipped at moderate severity before rising again, resulting in a slight overall rise in total DT. These findings suggest dynamic remodeling under sustained hemodynamic load, with potential implications for impaired retinal autoregulation and heightened systemic vascular risk (Table 5).

In hyperlipidemia, progressive microvascular impairment was similarly evident. CRAE showed a modest but significant increase from 181.7 μm to 188.6 μm (p-trend = 0.015), possibly reflecting compensatory arteriolar remodeling in response to lipid-induced endothelial stress. CRVE increased significantly from 297.6 μm to

Table 5: Trends in Retinal Features Across Different Severity Groups Stratified by Presence of Hypertension[†]

Retinal Feature	Low	Moderate	Severe	ANOVA p-value	p-trend
CRAE	182.674 $\pm$ 22.204	183.346 $\pm$ 21.071	188.048 $\pm$ 19.095	0.099	0.052
CRVE	297.397 $\pm$ 34.474	306.087 $\pm$ 35.992	312.117 $\pm$ 32.790	0.004	0.001
AVR	0.617 $\pm$ 0.056	0.603 $\pm$ 0.060	0.606 $\pm$ 0.060	0.125	0.134
FD	1.341 $\pm$ 0.041	1.347 $\pm$ 0.042	1.356 $\pm$ 0.038	0.023	0.007
VD	0.031 $\pm$ 0.007	0.031 $\pm$ 0.007	0.033 $\pm$ 0.006	0.014	0.005
Artery FD	1.292 $\pm$ 0.035	1.292 $\pm$ 0.035	1.296 $\pm$ 0.031	0.533	0.313
Vein FD	1.321 $\pm$ 0.034	1.325 $\pm$ 0.033	1.334 $\pm$ 0.029	0.007	0.002
Artery DT	3.725 $\pm$ 2.099	3.984 $\pm$ 2.038	4.067 $\pm$ 2.180	0.401	0.196
Vein DT	3.680 $\pm$ 1.781	3.533 $\pm$ 1.673	3.843 $\pm$ 1.821	0.373	0.512
DT	3.212 $\pm$ 1.088	3.237 $\pm$ 1.024	3.319 $\pm$ 1.036	0.713	0.434

314.5 μm (p-trend = 0.000), consistent with venular dilation observed in systemic inflammatory states. Although AVR remained relatively stable, fractal dimension rose from 1.34 to 1.36 (p-trend = 0.005), and vessel density increased from 0.031 to 0.033 (p-trend = 0.002), indicating subtle changes in capillary complexity and perfusion (Table 6). These retinal changes are indicative of systemic endothelial dysfunction promoted by lipid accumulation and inflammation, which are central to atherogenesis and contribute to increased cardiovascular risk.

Table 6: Trends in Retinal Features Across Different Severity Groups Stratified by Presence of Hyperlipidemia[†]

Retinal Feature	Low	Moderate	Severe	ANOVA p-value	p-trend
CRAE	181.731 $\pm$ 22.777	184.935 $\pm$ 20.276	188.591 $\pm$ 18.973	0.052	0.015
CRVE	297.607 $\pm$ 36.564	302.676 $\pm$ 34.005	314.542 $\pm$ 32.561	0.001	0.000
AVR	0.614 $\pm$ 0.053	0.615 $\pm$ 0.060	0.604 $\pm$ 0.064	0.332	0.268
FD	1.341 $\pm$ 0.041	1.349 $\pm$ 0.040	1.355 $\pm$ 0.039	0.020	0.005
VD	0.031 $\pm$ 0.007	0.032 $\pm$ 0.007	0.033 $\pm$ 0.007	0.008	0.002
Artery FD	1.292 $\pm$ 0.035	1.298 $\pm$ 0.034	1.297 $\pm$ 0.031	0.357	0.225
Vein FD	1.320 $\pm$ 0.035	1.325 $\pm$ 0.034	1.335 $\pm$ 0.028	0.003	0.001
Artery DT	3.796 $\pm$ 2.116	4.188 $\pm$ 2.091	4.054 $\pm$ 2.245	0.303	0.289
Vein DT	3.526 $\pm$ 1.627	3.457 $\pm$ 1.704	3.967 $\pm$ 1.803	0.068	0.086
DT	3.160 $\pm$ 1.063	3.307 $\pm$ 1.073	3.362 $\pm$ 1.022	0.304	0.136

Retinal vascular changes in kidney disease demonstrate complex microvascular remodeling linked to renal microangiopathy and systemic endothelial dysfunction. CRAE steadily increased from about 176.0 μm in low severity group to nearly 190.0 μm in severe cases (p-trend = 0.0284), likely reflecting arteriolar dilation as a compensatory response to chronic ischemia and hypoxia. AVR remained relatively stable between low and moderate severity (0.61), but showed a noticeable increase in severe disease (0.63), suggesting some disproportionate remodeling favoring arterioles over venules in advanced stages. Noteworthy increases were observed in fractal dimension, rising from 1.33 to 1.36 (p-trend = 0.009), and vessel density, which climbed from 0.028 to 0.034 (p-trend = 0.008). These trends indicate enhanced vascular complexity, potentially driven by adaptive angiogenesis or increased capillary recruitment as the disease advances. Arterial fractal dimension similarly rose from 1.28 to 1.31 (p-trend = 0.009), further supporting ongoing structural vascular changes. Altogether, these findings highlight the intricate and evolving remodeling of retinal microvasculature in kidney disease, balancing vascular injury and compensatory adaptations under persistent hemodynamic and metabolic stress (Table 7).

Among patients with cardiovascular disease, retinal vascular changes corresponded with increasing physiological severity. CRVE steadily increased from 299.3 μm to 313.5 μm (p-trend = 0.039), indicating possible venous pressure elevation or vascular congestion.

Table 7: Trends in Retinal Features Across Different Severity Groups Stratified by Presence of Kidney Disease[†]

Retinal Feature	Low	Moderate	Severe	ANOVA p-value	p-trend
CRAE	175.960 ± 21.564	183.599 ± 21.997	189.992 ± 17.012	0.092	0.028
CRVE	286.924 ± 25.166	301.634 ± 38.401	301.427 ± 25.864	0.178	0.107
AVR	0.615 ± 0.056	0.613 ± 0.058	0.634 ± 0.060	0.463	0.326
FD	1.325 ± 0.043	1.345 ± 0.053	1.363 ± 0.043	0.035	0.009
VD	0.028 ± 0.006	0.031 ± 0.009	0.034 ± 0.008	0.031	0.008
Artery FD	1.281 ± 0.037	1.298 ± 0.033	1.308 ± 0.032	0.033	0.010
Vein FD	1.311 ± 0.036	1.322 ± 0.033	1.336 ± 0.032	0.063	0.018
Artery DT	3.655 ± 2.087	3.627 ± 1.974	3.365 ± 1.814	0.879	0.648
Vein DT	3.626 ± 2.112	3.644 ± 2.030	3.839 ± 1.961	0.937	0.747
DT	3.219 ± 0.890	2.859 ± 0.917	3.238 ± 0.949	0.304	0.933

CRAE showed a slight, non-significant increase from 184.9 μ m to 192.5 μ m, suggesting variable arteriolar dilation. AVR remained stable across severity levels, implying proportional expansion of arterioles and venules or homeostatic mechanisms preserving balance. FD showed no significant changes, indicating that vascular complexity and perfusion density are largely maintained despite systemic cardiovascular stress. Notably, overall vessel tortuosity (DT) increased significantly from 3.08 to 3.48 (p-trend = 0.049), reflecting progressive retinal microvascular remodeling. Although arterial and venous tortuosity changes were inconsistent individually, the combined DT rise signals mounting microvascular strain. These vascular adaptations align with deteriorating physiological status, as those with lower physical activity or higher wearable-derived abnormality scores exhibit greater microvascular remodeling (Table 8).

Table 8: Trends in Retinal Features Across Different Severity Groups Stratified by Presence of Heart Disease[†]

Retinal Feature	Low	Moderate	Severe	ANOVA p-value	p-trend
CRAE	184.863 ± 20.462	185.619 ± 22.956	192.470 ± 24.289	0.198	0.111
CRVE	299.334 ± 30.299	303.970 ± 32.347	313.491 ± 38.201	0.107	0.039
AVR	0.620 ± 0.055	0.614 ± 0.070	0.618 ± 0.068	0.887	0.828
FD	1.341 ± 0.042	1.353 ± 0.038	1.347 ± 0.047	0.295	0.427
VD	0.031 ± 0.007	0.032 ± 0.007	0.033 ± 0.008	0.292	0.159
Artery FD	1.292 ± 0.037	1.298 ± 0.036	1.298 ± 0.036	0.618	0.415
Vein FD	1.321 ± 0.036	1.327 ± 0.032	1.334 ± 0.031	0.168	0.061
Artery DT	3.662 ± 1.874	4.435 ± 2.482	4.063 ± 2.261	0.161	0.285
Vein DT	3.540 ± 1.535	3.957 ± 2.136	3.726 ± 1.762	0.456	0.532
DT	3.080 ± 1.051	3.418 ± 1.077	3.482 ± 1.048	0.105	0.049

Neurological conditions, particularly those associated with neurodegeneration and small vessel disease, demonstrated significant venous dilation, with CRVE increasing from 290.7 μ m to 310.6 μ m (p-trend = 0.013), accompanied by progressive arteriolar narrowing as AVR declined from 0.62 to 0.59 (p-trend = 0.017). These vascular changes reflect microvascular remodeling and dysregulation characteristic of cerebral small vessel pathology, contributing to neurovascular impairment. Beyond statistically significant results, manual observations revealed gradual increases in vessel density (0.030 to 0.032) and arterial tortuosity (3.88 to 4.39), suggesting early compensatory or adaptive vascular responses, despite lacking formal significance. FD also showed a mild upward trend (1.333 to 1.349), indicating maintained or remodeled branching complexity. Overall, retinal microvascular alterations provide valuable insight into cerebral vascular health and the progression of neurological disease (Table 9).

This study demonstrates the complementary value of integrating retinal vascular imaging with wearable-derived physiological

Table 9: Trends in Retinal Features Across Different Severity Groups Stratified by Presence of Neuro Disease[†]

Retinal Feature	Low	Moderate	Severe	ANOVA p-value	p-trend
CRAE	180.417 ± 22.578	178.410 ± 18.174	182.705 ± 16.143	0.623	0.807
CRVE	290.716 ± 34.774	296.911 ± 33.201	310.554 ± 23.037	0.036	0.013
AVR	0.624 ± 0.057	0.606 ± 0.067	0.591 ± 0.052	0.058	0.017
FD	1.333 ± 0.044	1.341 ± 0.036	1.349 ± 0.042	0.205	0.074
VD	0.030 ± 0.007	0.030 ± 0.006	0.032 ± 0.006	0.305	0.152
Artery FD	1.292 ± 0.036	1.293 ± 0.034	1.295 ± 0.034	0.946	0.744
Vein FD	1.315 ± 0.035	1.318 ± 0.034	1.329 ± 0.030	0.178	0.086
Artery DT	3.880 ± 2.326	4.238 ± 2.303	4.387 ± 2.164	0.552	0.290
Vein DT	3.625 ± 1.803	3.945 ± 2.112	3.611 ± 1.226	0.568	0.795
DT	3.207 ± 1.111	3.484 ± 1.072	3.407 ± 0.802	0.322	0.259

data to assess systemic health. Among the retinal features analyzed, CRVE (venular caliber), AVR, vein fractal dimension, and vessel density showed the most consistent and statistically significant associations with increasing physiological severity, particularly within disease-specific subgroups such as diabetes, hypertension, and kidney disease. In contrast, CRAE and tortuosity measures did not exhibit significant or consistent trends across severity strata. These findings suggest that specific retinal biomarkers, especially those capturing venular remodeling and microvascular complexity can augment the interpretability of continuous physiological data from wearables. Together, this multimodal approach offers a scalable, non-invasive framework for stratifying systemic vascular risk and monitoring chronic disease progression.

5 Conclusion

This study demonstrates the potential of integrating retinal vascular imaging with continuous physiological data from wearable devices to enhance systemic disease assessment. By identifying specific retinal features, particularly those reflecting venular remodeling and microvascular complexity, that correlate with physiological severity, we propose a scalable and noninvasive framework for early detection, risk stratification, and monitoring of chronic diseases. This multimodal oculomic physiologic approach offers a promising avenue for advancing personalized medicine through more precise and timely interventions. However, the study is limited by a short monitoring period of 10 days, which may not fully capture long-term physiological variability or disease progression. Future longitudinal studies are essential to validate the reliability and clinical utility of these biomarkers. Additionally, incorporating optic disc related features and other retinal biomarkers could further enrich the understanding of ocular systemic relationships. In conclusion, the integration of retinal imaging with wearable data presents a compelling strategy for noninvasive, continuous, and individualized health monitoring, laying the foundation for improved disease management and preventive care.

[†]Note: Values are presented as mean ± standard deviation. CRAE: Central Retinal Arteriolar Equivalent; CRVE: Central Retinal Venular Equivalent; AVR: Arteriolar-to-Venular Ratio; FD: Fractal Dimension; VD: Vessel Density; DT: Distance Tortuosity. Significant p-values (p < 0.05) are highlighted in bold.

References

- [1] Salar Abbaspourazad, Oussama Elachqar, Andrew C Miller, Saba Emrani, Udhayakumar Nallasamy, and Ian Shapiro. 2023. Large-scale training of foundation models for wearable biosignals. (2023). arXiv:2312.05409 [cs.LG]
- [2] Brinnae Bent, Peter J Cho, Maria Henriquez, April Wittmann, Connie Thacker, Mark Feinglos, Matthew J Crowley, and Jessilyn P Dunn. 2021. Engineering digital biomarkers of interstitial glucose from noninvasive smartwatches. *NPJ Digit. Med.* 4, 1 (June 2021), 89.
- [3] Hangyoung Choi, Jaeseong Hong, Hyun Goo Kang, Min-Hyeon Park, Sungji Ha, Junghan Lee, Sangchul Yoon, Daeseong Kim, Yu Rang Park, and Keun-Ah Cheon. 2025. Retinal fundus imaging as biomarker for ADHD using machine learning for screening and visual attention stratification. *NPJ Digit. Med.* 8, 1 (March 2025), 164.
- [4] Jean-Francois Daneault, Gloria Vergara-Diaz, Federico Parisi, Chen Admati, Christina Alfonso, Matilde Bertoli, Edoardo Bonizzoni, Gabriela Ferreira Carvalho, Gianluca Costante, Eric Eduardo Fabara, Naama Fixler, Fatemah Noushin Golabchi, John Growdon, Stefano Sapienza, Phil Snyder, Shahar Shpigelman, Lewis Sudarsky, Margaret Daeschler, Lauren Bataille, Solveig K Sieberts, Larsson Omberg, Steven Moore, and Paolo Bonato. 2021. Accelerometer data collected with a minimum set of wearable sensors from subjects with Parkinson's disease. *Sci. Data* 8, 1 (Feb. 2021), 48.
- [5] Ciprian Danieleescu, Marius Gabriel Dabija, Alin Horatiu Nedelcu, Vasile Valeriu Lupu, Ancuta Lupu, Ileana Ioniuc, Georgiana-Emmanuela Gilcă-Blanariu, Vlad-Constantin Donica, Maria-Luciana Anton, and Ovidiu Musat. 2023. Automated retinal vessel analysis based on fundus photographs as a predictor for non-ophthalmic diseases-evolution and perspectives. *J. Pers. Med.* 14, 1 (Dec. 2023).
- [6] Aiden Doherty, Dan Jackson, Nils Hammerla, Thomas Plötz, Patrick Olivier, Malcolm H Granat, Tom White, Vincent T van Hees, Michael I Trenell, Christopher G Owen, Stephen J Preece, Rob Gillions, Simon Sheard, Tim Peakman, Soren Brage, and Nicholas J Wareham. 2017. Large scale population assessment of physical activity using wrist worn accelerometers: The UK biobank study. *PLoS One* 12, 2 (Feb. 2017), e0169649.
- [7] Pétra Eid, Abderrahmane Bourredjem, Atif Anwer, Catherine Creuzot-Garcher, Pearse Andrew Keane, Yukun Zhou, Siegfried Wagner, Fabrice Meriaudeau, and Louis Arnould. 2025. Retinal microvascular biomarker assessment with automated algorithm and semiautomated software in the Montrachet dataset. *Transl. Vis. Sci. Technol.* 14, 3 (March 2025), 13.
- [8] Nada El Hussein, Christopher L Schaich, Suzanne Craft, Stephen R Rapp, Kathleen M Hayden, Richey Sharrett, Mary Frances Cotch, Tien Y Wong, Jose A Luchsinger, Mark A Espeland, Laura D Baker, Alain G Bertoni, and Timothy M Hughes. 2024. Retinal vessel caliber and cognitive performance: the multi-ethnic study of atherosclerosis (MESA). *Sci. Rep.* 14, 1 (Feb. 2024), 4120.
- [9] Justin Engelmann, Diana Moukaddem, Lucas Gago, Niall Strang, and Miguel O Bernabeu. 2024. Applicability of oculomics for individual risk prediction: Repeatability and robustness of retinal Fractal Dimension using DART and AutoMorph. (2024). arXiv:2403.06950 [q-bio.QM]
- [10] M M Fraz, R A Welikala, A R Rudnicka, C G Owen, D P Strachan, and S A Barman. 2015. QUARTZ: Quantitative Analysis of Retinal Vessel Topology and size – An automated system for quantification of retinal vessels morphology. *Expert Syst. Appl.* 42, 20 (Nov. 2015), 7221–7234.
- [11] M M Fraz, R A Welikala, A R Rudnicka, C G Owen, D P Strachan, and S A Barman. 2015. QUARTZ: Quantitative Analysis of Retinal Vessel Topology and size – An automated system for quantification of retinal vessels morphology. *Expert Syst. Appl.* 42, 20 (Nov. 2015), 7221–7234.
- [12] Yuechuan Fu, Mayinuer Yusufu, Yueye Wang, Mingguang He, Danli Shi, and Ruobing Wang. 2023. Association of retinal microvascular density and complexity with incident coronary heart disease. *Atherosclerosis* 380 (Sept. 2023), 117196.
- [13] Joseph Futoma, Morgan Simons, Trishan Panch, Finale Doshi-Velez, and Leo Anthony Celi. 2020. The myth of generalisability in clinical research and machine learning in health care. *Lancet Digit. Health* 2, 9 (Sept. 2020), e489–e492.
- [14] Samuel D Giesser, Ferhat Turgut, Amr Saad, Jay R Zoellin, Chiara Sommer, Yukun Zhou, Siegfried K Wagner, Pearse A Keane, Matthias Becker, Delia Cabrera DeBuc, and Gábor Márk Somfai. 2024. Evaluating the impact of retinal vessel segmentation metrics on retest reliability in a clinical setting: A comparative analysis using AutoMorph. *Invest. Ophthalmol. Vis. Sci.* 65, 13 (Nov. 2024), 24.
- [15] Fan Huang, Behdad Dashtbozorg, and Bart M Ter Haar Romeny. 2018. Artery/vein classification using reflection features in retina fundus images. *Mach. Vis. Appl.* 29, 1 (Jan. 2018), 23–34.
- [16] Andrew Hughes, Md Mobashir Hasan Shandhi, Hiral Master, Jessilyn Dunn, and Evan Brittain. 2023. Wearable devices in cardiovascular medicine. *Circ. Res.* 132, 5 (March 2023), 652–670.
- [17] Ryo Kawasaki, Ning Cheung, Jie Jin Wang, Ronald Klein, Barbara E K Klein, Mary Frances Cotch, A Richey Sharrett, Steven Shea, F M Amirul Islam, and Tien Y Wong. 2009. Retinal vessel diameters and risk of hypertension: the Multiethnic Study of Atherosclerosis. *J. Hypertens.* 27, 12 (Dec. 2009), 2386–2393.
- [18] Michael D Knudston, Kristine E Lee, Larry D Hubbard, Tien Yin Wong, Ronald Klein, and Barbara E K Klein. 2003. Revised formulas for summarizing retinal vessel diameters. *Curr. Eye Res.* 27, 3 (Sept. 2003), 143–149.
- [19] Sophie Lemmens, Astrid Devulder, Karel Van Keer, Johan Bierkens, Patrick De Boever, and Ingeborg Stalmans. 2020. Systematic review on fractal dimension of the retinal vasculature in neurodegeneration and stroke: Assessment of a potential biomarker. *Front. Neurosci.* 14 (Jan. 2020), 16.
- [20] Gerald Liew, Bamini Gopinath, Andrew J White, George Burlutsky, Tien Yin Wong, and Paul Mitchell. 2021. Retinal vasculature fractal and stroke mortality. *Stroke* 52, 4 (April 2021), 1276–1282.
- [21] Gerald Liew, Bamini Gopinath, Andrew J White, George Burlutsky, Tien Yin Wong, and Paul Mitchell. 2021. Retinal vasculature fractal and stroke mortality. *Stroke* 52, 4 (April 2021), 1276–1282.
- [22] Gustav Mårtensson, Daniel Ferreira, Tobias Granberg, Lena Cavallin, Ketil Oppedal, Alessandro Padovani, Irena Rektorova, Laura Bonanni, Matteo Pardini, Milica G Kramberger, John-Paul Taylor, Jakub Hort, Jón Snædal, Jaime Kulisevsky, Frederic Blanc, Angelo Antonini, Patrizia Mecocci, Bruno Vellas, Magda Tsolaki, Iwona Kloszewska, Hilka Soininen, Simon Lovestone, Andrew Simmons, Dag Aarsland, and Eric Westman. 2020. The reliability of a deep learning model in clinical out-of-distribution MRI data: A multicohort study. *Med. Image Anal.* 66, 101714 (Dec. 2020), 101714.
- [23] National Heart, Lung, and Blood Institute. 2024. Atherosclerosis Risk in Communities (ARIC) Study. <https://www.nhlbi.nih.gov/science/atherosclerosis-risk-communities-aric-study> Accessed: 2025-04-30.
- [24] Niall Patton, Tariq Aslam, Thomas Macgillivray, Alison Pattie, Ian J Deary, and Baljean Dhillon. 2005. Retinal vascular image analysis as a potential screening tool for cerebrovascular disease: a rationale based on homology between cerebral and retinal microvasculatures. *J. Anat.* 206, 4 (April 2005), 319–348.
- [25] A Perez-Rovira, T MacGillivray, E Trucco, K S Chin, K Zutis, C Lupascu, D Tegolo, A Giachetti, P J Wilson, A Doney, and B Dhillon. 2011. VAMPIRE: Vessel assessment and measurement platform for images of the REtina. In *2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society* (Boston, MA). IEEE.
- [26] A Perez-Rovira, T MacGillivray, E Trucco, K S Chin, K Zutis, C Lupascu, D Tegolo, A Giachetti, P J Wilson, A Doney, and B Dhillon. 2011. VAMPIRE: Vessel assessment and measurement platform for images of the REtina. In *2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society* (Boston, MA). IEEE.
- [27] Lukasz Piwek, David A Ellis, Sally Andrews, and Adam Joinson. 2016. The rise of consumer health wearables: Promises and barriers. *PLoS Med.* 13, 2 (Feb. 2016), e1001953.
- [28] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Lecture Notes in Computer Science*. Springer International Publishing, Cham, 234–241.
- [29] Emmanuel Sandoval-Garcia, Stela McLachlan, Anna H Price, Thomas J MacGillivray, Mark W J Strachan, James F Wilson, and Jackie F Price. 2021. Retinal arteriolar tortuosity and fractal dimension are associated with long-term cardiovascular outcomes in people with type 2 diabetes. *Diabetologia* 64, 10 (Oct. 2021), 2215–2227.
- [30] Muhammad Bayu Sasongko, Tien Yin Wong, Kim C Donaghue, Ning Cheung, Alicia J Jenkins, Paul Benitez-Aguirre, and Jie Jin Wang. 2012. Retinal arteriolar tortuosity is associated with retinopathy and early kidney dysfunction in type 1 diabetes. *Am. J. Ophthalmol.* 153, 1 (Jan. 2012), 176–83.e1.
- [31] Shaghayegh Shajari, Kirankumar Kuruvinashetti, Amin Komeili, and Uttandaraman Sundararaj. 2023. The emergence of AI-based wearable sensors for digital health technology: A review. *Sensors (Basel)* 23, 23 (Nov. 2023).
- [32] Matthew Smuck, Charles A Odonkor, Jonathan K Wilt, Nicolas Schmidt, and Michael A Swiernik. 2021. The emerging clinical role of wearables: factors for successful implementation in healthcare. *NPJ Digit. Med.* 4, 1 (March 2021), 45.
- [33] Chetan L Srinidhi, Aparna P, and Jeny Rajan. 2019. Automated method for retinal artery/vein separation via graph search metaheuristic approach. *IEEE Trans. Image Process.* 28, 6 (Jan. 2019), 2705–2718.
- [34] Lie Ming Tang, Jochen Meyer, Daniel A Epstein, Kevin Bragg, Lina Engelen, Adrian Bauman, and Judy Kay. 2018. Defining adherence. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 2, 1 (March 2018), 1–22.
- [35] James Truslow, Angela Spillane, Huiming Lin, Katherine Cyr, Adeeti Ullal, Edith Arnold, Ron Huang, Laura Rhodes, Jennifer Block, Jamie Stark, et al. 2024. Understanding activity and physiology at scale: The Apple Heart & Movement Study. *npj Digital Medicine* 7, 1 (2024), 242.
- [36] Ana Villaplana-Velasco, Marie Pigeyre, Justin Engelmann, Konrad Rawlik, Oriol Canela-Xandri, Claire Tochel, Frida Lona-Durazo, Muthu Rama Krishnan Mookiah, Alex Doney, Esteban J Parra, Emanuele Trucco, Tom MacGillivray, Kristiina Rannikmaa, Albert Tenesa, Erola Pairo-Castineira, and Miguel O Bernabeu. 2023. Fine-mapping of retinal vascular complexity loci identifies Notch regulation as a shared mechanism with myocardial infarction outcomes. *Commun. Biol.* 6, 1 (May 2023), 523.
- [37] Siegfried K Wagner, Dun Jack Fu, Livia Faes, Xiaoxuan Liu, Josef Huemer, Hagar Khalid, Daniel Ferraz, Edward Korot, Christopher Kelly, Konstantinos Balaskas, Alastair K Denniston, and Pearse A Keane. 2020. Insights into systemic disease through retinal imaging-based oculomics. *Transl. Vis. Sci. Technol.* 9, 2 (Feb. 2020),

- 6.
- [38] Siegfried K Wagner, Dun Jack Fu, Livia Faes, Xiaoxuan Liu, Josef Huemer, Hagar Khalid, Daniel Ferraz, Edward Korot, Christopher Kelly, Konstantinos Balaskas, Alastair K Denniston, and Pearse A Keane. 2020. Insights into systemic disease through retinal imaging-based oculomics. *Transl. Vis. Sci. Technol.* 9, 2 (Feb. 2020), 6.
- [39] Tien Yin Wong, F M Amirul Islam, Ronald Klein, Barbara E K Klein, Mary Frances Cotch, Cecilia Castro, A Richey Sharrett, and Eyal Shahar. 2006. Retinal vascular caliber, cardiovascular risk factors, and inflammation: the multi-ethnic study of atherosclerosis (MESA). *Invest. Ophthalmol. Vis. Sci.* 47, 6 (June 2006), 2341–2350.
- [40] T Y Wong, R Klein, D J Couper, L S Cooper, E Shahar, L D Hubbard, M R Wofford, and A R Sharrett. 2001. Retinal microvascular abnormalities and incident stroke: the Atherosclerosis Risk in Communities Study. *Lancet* 358, 9288 (Oct. 2001), 1134–1140.
- [41] Tien Yin Wong, Ronald Klein, A Richey Sharrett, Bruce B Duncan, David J Couper, James M Tielsch, Barbara E K Klein, and Larry D Hubbard. 2002. Retinal arteriolar narrowing and risk of coronary heart disease in men and women. *JAMA* 287, 9 (March 2002).
- [42] Linbin Wu, Xia Gong, Wei Wang, Lei Zhang, Jiachen Zhou, Xi Ming, Meng Yuan, Wenying Huang, and Lanhua Wang. 2022. Association of retinal fractal dimension and vessel tortuosity with impaired renal function among healthy Chinese adults. *Front. Med. (Lausanne)* 9 (Sept. 2022), 925756.
- [43] Hang Yuan, Shing Chan, Andrew P Creagh, Catherine Tong, Aidan Acquah, David A Clifton, and Aiden Doherty. 2024. Self-supervised learning for human activity recognition using 700,000 person-days of wearable data. *NPJ Digit. Med.* 7, 1 (April 2024), 91.
- [44] Yukun Zhou, Zailiang Chen, Hailan Shen, Xianxian Zheng, Rongchang Zhao, and Xuanchu Duan. 2021. A refined equilibrium generative adversarial network for retinal vessel segmentation. *Neurocomputing* 437 (May 2021), 118–130.
- [45] Yukun Zhou, Mark A Chia, Siegfried K Wagner, Murat S Ayhan, Dominic J Williamson, Robbert R Struyven, Timing Liu, Moucheng Xu, Mateo G Lozano, Peter Woodward-Court, Yuka Kihara, UK Biobank Eye & Vision Consortium, Andre Altmann, Aaron Y Lee, Eric J Topol, Alastair K Denniston, Daniel C Alexander, and Pearse A Keane. 2023. A foundation model for generalizable disease detection from retinal images. *Nature* 622, 7981 (Oct. 2023), 156–163.
- [46] Yukun Zhou, Siegfried K Wagner, Mark A Chia, An Zhao, Peter Woodward-Court, Moucheng Xu, Robbert Struyven, Daniel C Alexander, and Pearse A Keane. 2022. AutoMorph: Automated retinal vascular morphology quantification via a deep learning pipeline. *Transl. Vis. Sci. Technol.* 11, 7 (July 2022), 12.